V2I Traffic Sim

by DeepCyan

INTRODUCTION

Smart cities need intelligent traffic systems. Traditional traffic lights are inefficient and cannot adapt to real-time conditions.

Our project presents an AI-powered V2I simulation platform that improves traffic flow, reduces congestion, and enhances road safety.

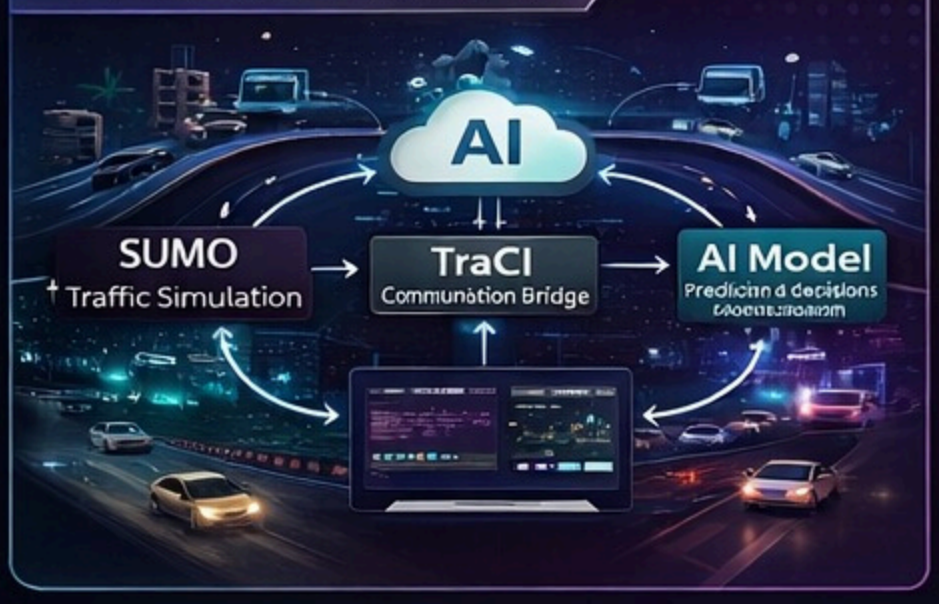
PROBLEM

- Fixed traffic signals cause delays
- Increasing urban congestion
- No real-time adaptability
- Expensive real-world testing

SOLUTION

- We developed a virtual AI-driven **Simulation platform** that:
- Simulates vehicle-to-infrastructure (V2I) communication
- Uses Machine Learning (Q-Learning & Deep Q-Learning)
- Predicts congestion and collision risks
- Dynamically adjusts traffic signals

SYSTEM OVERVIEW



RESULTS / IMPACT

- Reduced traffic congestion
- Improved traffic flow efficiency
- Safer road environment
- Cost-effective testing platform

KEY FEATURES

- Real-time traffic simulation
- AI-based congestion prediction
- Collision detection system
- Interactive dashboard

TECHNOLOGIES USED

- Python
- SUMO
- FastAPI
- TensorFlow
- React.js Dashboard
- Docker

TEAM

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CONCLUSION

This project demonstrates how **AI + V2I communication** can transform future transportation systems and smart cities.